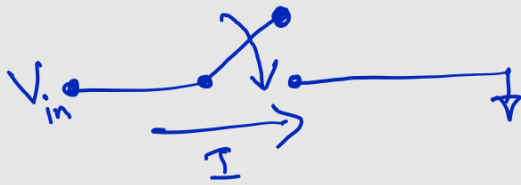
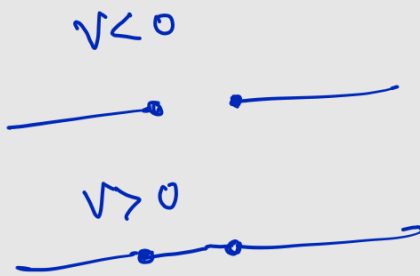
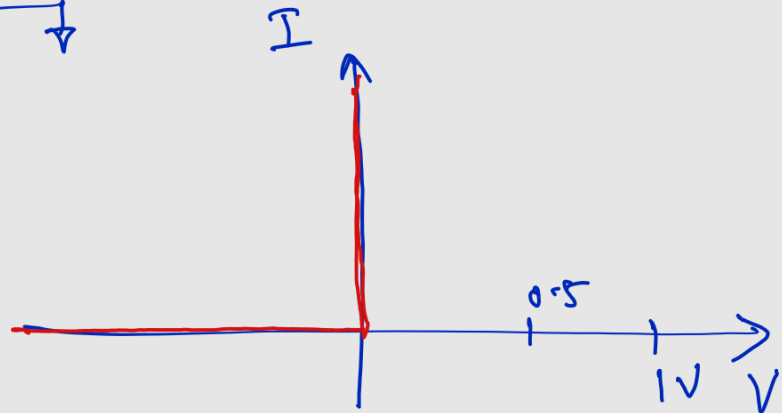


* Switch : (I-V)

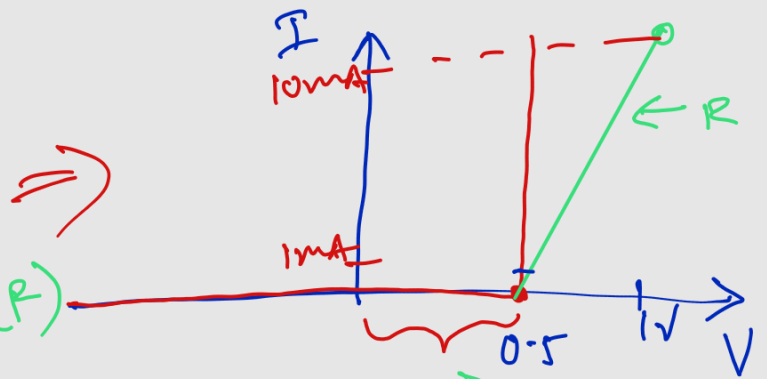


$$I \quad V_{in} > 0$$

$$0 \quad V_{in} < 0$$



Some resistance (R)
offered by wire



Shift to consider
voltage tolerance

$$I = 0 \quad V \leq 0.5$$

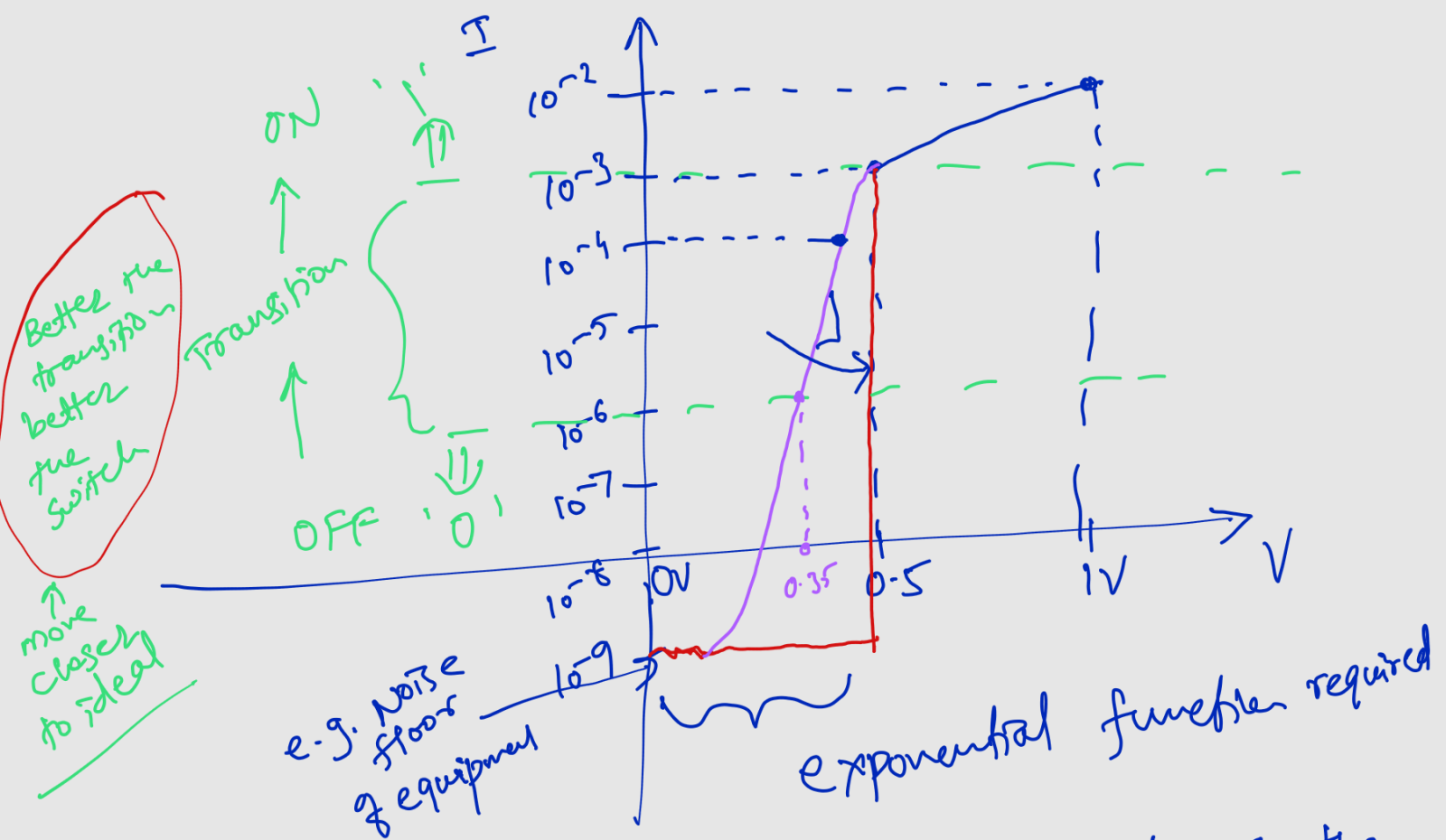
$$= \frac{V - 0.5}{R} \quad V > 0.5$$

Some number
for example

$$V = 0.51 \text{ V}, \quad I = 1 \text{ mA}$$

$$V = 0.49 \text{ V}, \quad I = 1 \text{ nA} \quad (\text{very low current})$$

Q can we plot in log scale?



* How much voltage is required to change the current 10 times? $\Rightarrow \frac{mV}{dec}$

we need to make a device which has good (close to ideal) function

e.g. "diode" / ~~not~~

$$I = A \cdot e^{B \cdot V}$$

$$= \frac{V - 0.5}{R}$$

$V \leq 0.5V$

$V > 0.5V$

V_{bi} not fixed